Linear Algebra – MAT 2610

Section 5.3 (Diagonalization)

Dr. Jay Adamsson

jay@aorweb.ca

jadamsson@upei.ca

Recall the following problem:

Suppose that in any given year in the city of Metropolis, 15% of the city dwellers move to the suburbs and 10% of the suburbanites move to the city. Also suppose there are no other changes in the population.

Define the variables:

c_0 is the city population at time 0, c_n is the population in year n
s_0 is the suburb population at time 0 , s_n is the population in year n

$$\begin{bmatrix} c_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

$$\begin{bmatrix} c_{k+1} \\ s_{k+1} \end{bmatrix} = \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix} \begin{bmatrix} c_k \\ s_k \end{bmatrix}$$

What about the population at year n? If we only know the current population at time 0, we can get an equation for the population n years in the future.

$$\begin{bmatrix} c_n \\ s_n \end{bmatrix} = \left(\begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix} \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix} \cdots \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix} \right) \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

$$\begin{bmatrix} c_n \\ s_n \end{bmatrix} = \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix}^n \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

How can we do this easily for large values of n? What is the population after 5 years?

Definition

An $n \times n$ matrix A is **diagonalizable** if there is an $n \times n$ matrix D such that $\underbrace{A = PDP^{-1}}_{3 \times 3}$ for some invertible matrix P

In other words, A is similar to D

Note: If $A = PDP^{-1}$ then $A^n = PD^nP^{-1}$

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \qquad D^m = \begin{bmatrix} \lambda_1^m & & 0 \\ & \ddots & \\ 0 & & \lambda_n^m \end{bmatrix}$$

Theorem 5 – The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $\cancel{a}\,A = P^{-1}DP$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

Example:

Let $A = \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix}$. Find a diagonal matrix D and invertible matrix P with $A = P^{-1}DP$

To get D, we need the eigenvalues of A

$$\det \begin{bmatrix} .85-t & .10 \\ .15 & .9-t \end{bmatrix} = (.85-t)(.9-t) - (.15)(.10) = 0$$

$$= .765 - 1.75t + t^2 - 0.015 = 0$$

$$t^2 - 1.75t + 0.75 = 0$$

$$(t-1)(t-0.75) = 0$$

eigenvalues: $\lambda_1 = 1$

$\lambda_2 = 0.75$

so $D = \begin{bmatrix} 1 & 0 \\ 0 & 0.75 \end{bmatrix}$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 0.75 \end{bmatrix}$$

To get D, find the eigenvectors

$$\lambda = 1: \quad \begin{bmatrix} -.15 & .10 \\ .15 & -0.1 \end{bmatrix} \longrightarrow \begin{bmatrix} -0.15 & 0.1 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -2/3 \\ 0 & 0 \end{bmatrix}$$

$$x_1 - \frac{2}{3}x_2 = 0 \qquad \Rightarrow \qquad x_1 = -\frac{2}{3}x_2 \qquad \text{Basis } \left\{ \begin{bmatrix} -2/3 \\ 1 \end{bmatrix} \right\}$$

$$x_2 = x_2 \qquad\qquad x_2 = x_2$$

$$\lambda = 0.75: \quad \begin{bmatrix} .10 & .10 \\ .15 & .15 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \qquad \text{Basis } \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

$$P = \begin{bmatrix} -2/3 & -1 \\ 1 & 1 \end{bmatrix}$$

$$A = PDP^{-1}$$

$$AP = PD$$

Returning to the problem we started with, what is the population after 5 years?

$$\begin{bmatrix} c_5 \\ s_5 \end{bmatrix} = \begin{bmatrix} .85 & .10 \\ .15 & .90 \end{bmatrix}^5 \begin{bmatrix} c_0 \\ s_0 \end{bmatrix} = PD^5P^{-1} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

$$\begin{bmatrix} c_5 \\ s_5 \end{bmatrix} = \begin{bmatrix} 2/3 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3/4 \end{bmatrix}^5 \begin{bmatrix} 3/5 & 3/5 \\ -3/5 & 2/5 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

$$\begin{bmatrix} c_5 \\ s_5 \end{bmatrix} = \begin{bmatrix} 2/3 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 243/1024 \end{bmatrix} \begin{bmatrix} 3/5 & 3/5 \\ -3/5 & 2/5 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

$$\begin{bmatrix} c_5 \\ s_5 \end{bmatrix} = \begin{bmatrix} 0.542 & 0.305 \\ 0.458 & 0.695 \end{bmatrix} \begin{bmatrix} c_0 \\ s_0 \end{bmatrix}$$

Theorem 5 – Proof

Let P be an $n \times n$ matrix with columns $v_1, \ldots, v_n$ and D a diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$.

$$AP = A[v_1\ v_2\ \ldots v_n] = [Av_1\ Av_2\ \ldots Av_n]$$

Also

$$PD = P\begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} = [\lambda_1 v_1\ \ldots \lambda_n v_n]$$

Theorem 5 – Proof

(1) Suppose $A = PDP^{-1}$, then $AP = PD$ and

$$\underbrace{[Av_1\ Av_2\ \ldots Av_n]} = \underbrace{[\lambda_1 v_1\ \ldots \lambda_n v_n]}$$

So

$$\boxed{Av_1 = \lambda_1 v_1}, \ldots, \boxed{Av_n = \lambda_n v_n}$$

Since P is invertible, its columns are linearly independent and are the eigenvectors that correspond to the eigenvalues.

Theorem 5 – Proof

(2) If A is an $n \times n$ matrix with eigenvectors $v_1, \ldots, v_n$, define

$\mathrm{P} = [v_1 \ v_2 \ \ldots v_n]$ and $D = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}$ where the

eigenvalue λ_i corresponds to the eigenvector v_i. We have already seen that the eigenvectors are linearly independent, so P is invertible. From the equations at the start of the proof, we can see that $AP = PD$, so $A = PDP^{-1}$

Theorem 6

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Note: An $n \times n$ matrix may be diagonalizable even if it does not have n distinct eigenvalues

Theorem 7(a)

If λ is an eigenvalue of an $n \times n$ then the dimension of the corresponding eigenspace is less than or equal to the multiplicity of λ.

Example

$\lambda = 6$ mult 1
$\lambda = -2$ mult 2

If $A = \begin{bmatrix} 6 & -4 & -4 \\ -8 & 2 & 4 \\ 8 & -4 & -6 \end{bmatrix}$ and its characteristic polynomial is $-(t-6)(t+2)^2$, then:

a. The eigenspace corresponding to the eigenvalue $\lambda = 6$ is equal to 1
b. The eigenspace corresponding to the eigenvalue $\lambda = -2$ is equal to either 1 or 2

Theorem 7(b)

An $n \times n$ matrix A is diagonalizable if and only if both of the conditions are true:

1. The total number of eigenvalues of A, when each eigenvalue is counted as often as its multiplicity, is equal to n; and
2. For each eigenvalue λ of A, the dimension of the corresponding eigenspace, which is $n - rank(A - \lambda I_n)$ is equal to the multiplicity of λ.

Example: Is $A = \begin{bmatrix} 6 & -4 & -4 \\ -8 & 2 & 4 \\ 8 & -4 & -6 \end{bmatrix}$ diagonalizable? If so, find D and P.

$\lambda_1 = 6$ mult 1

$\lambda_2 = -2$ mult 2

3 ✓ 1st condition

Hint: the characteristic polynomial is $-(t-6)(t+2)^2$

$$\lambda = 6 : \begin{bmatrix} 0 & -4 & -4 \\ -8 & -4 & 4 \\ 8 & -4 & -12 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Basis} = \left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$\lambda = -2$:
$$\begin{bmatrix} 8 & -4 & -4 \\ -8 & 4 & 4 \\ 8 & -4 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Basis} = \left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

This shows it is diagonalizable

$$D = \begin{bmatrix} 6 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \qquad P = \begin{bmatrix} 0 & 1/2 & 1/2 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Example: Is $A = \underbrace{\begin{bmatrix} 3 & -5 & 6 \\ 1 & 3 & -6 \\ 0 & 3 & -5 \end{bmatrix}}$ diagonalizable? If so, find D and P.

Hint: the characteristic polynomial is $\underbrace{-(t-1)(t^2+2)}$

$\lambda = 1$

no real solution

Not diagonalizable.

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